

Complex Hermitian Optimization and Spectral Alignment for Joint Qubit Admission

Benjamin Walker Mayes

Abstract—We formulate and execute a complex Hermitian optimization for a still-open six-qubit joint admission interface. A fixed planar graph permits 108 nonidentity two-local Hermitian Pauli coefficients. Eleven exact first-moment constraints leave 97 real coordinates, decomposing into 55 real-entry and 42 imaginary-entry directions. The objective retains all four input columns and one common phase; its analytic derivative uses conjugate-transpose spectral calculus with a continuous divided difference. All 392 numerical Jacobian-column checks and two directional checks pass. Eight bounded fits supply no encoder. The best sampled endpoint residual is 0.329104, not a channel certificate. An exact counterexample shows that terminal activation and all compressed spectral moments remain insufficient. Spectral-projector orientation supplies a sharper criterion. Exact dyadic backward-error enclosures certify an all-time half-diamond error greater than $1/8$ for each best frozen fit, using a maximally entangled logical/reference witness. These exclusions concern individual generators, not the planar family. The joint endpoint and physical realization remain open.

Index Terms—Autonomous control, Hermitian Hamiltonian, two-local interaction, complex gradient, quantum admission, spectral moment.

I. OPERATIONAL CONTRACT AND TYPED SEARCH SPACE

With register order (A, O, u, v, w, z) , define isometries

$$E|ab\rangle = |ab\rangle_{AO}|\mathbf{1}_{00}\rangle_M, \quad W|ab\rangle = |00\rangle_{AO}|\mathbf{1}_{ab}\rangle_M, \quad (1)$$

where modes (u, v, w, z) are labeled $(00, 01, 10, 11)$. We seek a static Hermitian two-local generator whose propagator maps E to W with one common phase, including arbitrary spectator entanglement. The full physical space has dimension 64. All candidate evolutions are unitary; failed endpoint admission is not evidence of nonunitary dynamics.

The permitted pairs remain

$$\{Au, Aw, Az, Ou, Ov, Ow, Oz, uw, uv, wz\}, \quad (2)$$

a planar graph with maximum degree four. The prior note [1] supplies the exact linear first-moment kernel. This note executes that previously unrun larger search. Bookkeeping charges Q_A, Q_O remain explicitly defined algebraic operators; no Gauss-law, Wilson-loop, or gravitational correspondence is derived. Static gate synthesis has prior spectral and numerical frameworks [2]; the gradient formulation itself is not asserted as a new general method.

There are $3 \cdot 6 + 9 \cdot 10 = 108$ nonidentity Hermitian Pauli strings P_k . All coefficients c_k are real. Odd numbers of Y factors permit imaginary matrix entries while preserving

MQM development note v0.1, October 5, 2026. ORCID: 0009-0002-5813-0724. Scoped derivation and finite constructor search. Independent review and theorem-level novelty comparison pending. Physical evidence promotion: 0.

Hermiticity. In particular Y is a one-body term; a direction of support at most two need not be exclusively two-body. First-moment alignment

$$E^\dagger H E = W^\dagger H W \quad (3)$$

has real rank eleven. An exact integer matrix $B \in \mathbb{Z}^{108 \times 97}$ parametrizes $c = B\theta$. The 62 real-entry Pauli terms have constraint rank seven and kernel dimension 55. The 46 imaginary-entry terms have rank four and kernel dimension 42. Thus the enlarged kernel is $55 + 42 = 97$, not the earlier 55-coordinate kernel. More independently tunable components are admitted, even though no qubits or allowed pairs are added.

II. OBJECTIVE WITH A COMMON PHASE

Put $F_a = \sum_k B_{ka} P_k$ and $H(\theta) = \sum_a \theta_a F_a$. We use dimensionless model time one and fit $\theta \in \mathbb{R}^{97}$ and $\phi \in \mathbb{R}$:

$$R = e^{-iH(\theta)} E - e^{i\phi} W, \quad f_0 = \frac{1}{2} \|R\|_F^2. \quad (4)$$

The residual stacks real and imaginary entries of all four columns. Each input is retained; independent column phases are not allowed. The phase derivative is

$$\partial_\phi R = -ie^{i\phi} W. \quad (5)$$

The endpoint operator residual $\|R\|$ gives an isometry-based upper bound on half-diamond channel distance, but its computed floating value is not an enclosed channel certificate. A large operator residual alone is not a lower bound; basis-state trace diagnostics and exact moment witnesses provide the failures reported below.

For the optional second-moment penalty, set

$$M_2 = E^\dagger H^2 E - W^\dagger H^2 W, \quad s = 1 + \sum_k c_k^2, \quad (6)$$

and minimize

$$f_\mu = \frac{1}{2} \|R\|_F^2 + \frac{\mu^2}{2} \|M_2/s\|_F^2. \quad (7)$$

We use $\mu = 0$ or 2. The penalty is soft; higher moments are necessary diagnostics, not exactly enforced nonlinear constraints. Exact first moment is preserved by freezing binary-rational kernel coordinates before coefficient assembly.

III. COMPLEX ANALYTIC GRADIENT

For $H = V\Lambda V^\dagger$, let $A_a = V^\dagger F_a V$ and $Q = V^\dagger E$. The exponential Fréchet derivative is

$$\partial_a (e^{-iH} E) = V(D \odot A_a)Q, \quad (8)$$

TABLE I
SAMPLED DIAGNOSTICS; NEITHER BATCH SUPPLIES AN ENCODER

Batch	Best operator residual	Spectral radius
Endpoint only	0.394308	40.622549
Second-moment penalty	0.329104	58.317288

where

$$D_{rs} = -ie^{-i(\lambda_r + \lambda_s)/2} \text{sinc}((\lambda_r - \lambda_s)/2), \quad (9)$$

and $\text{sinc } x = \sin x/x$, $\text{sinc } 0 = 1$. At coincidence $D_{rr} = -ie^{-i\lambda_r}$; no division by a small eigenvalue gap is required. Conjugate transposes cannot be replaced by ordinary transposes for this complex ansatz.

Derivative identity. Differentiation of the propagator gives

$$-i \int_0^1 e^{-i(1-q)H} F_a e^{-iqH} dq. \quad (10)$$

In the eigenbasis, integration of the two scalar phases yields the displayed continuous divided difference, including equal eigenvalues. Multiplication by E gives the stated formula. $\square$

For the penalty,

$$\partial_a M_2 = E^\dagger (F_a H + H F_a) E - W^\dagger (F_a H + H F_a) W, \quad (11)$$

$$\partial_a s = 2 \sum_k c_k B_{ka}, \quad (12)$$

$$\partial_a (M_2/s) = (\partial_a M_2)/s - M_2(\partial_a s)/s^2. \quad (13)$$

The loss gradient takes real Frobenius inner products with these derivatives. Equivalently, the solver uses the Jacobian of the stacked real residual. Four complete 98-column central-difference comparisons and two larger-spectrum directional comparisons pass, with maximum discrepancy below 10^{-8} ; these are numerical consistency checks, not a rigorous floating exponential enclosure.

IV. BOUNDED FITS AND EXACT FROZEN-CANDIDATE REJECTION

Each batch has four deterministic starts: the preceding best embedded fit, a seed perturbed in imaginary-entry directions, and two seeded random vectors. Each start has at most 1000 function evaluations and a coordinate box $[-4\pi, 4\pi]$. This box is not a Pauli coefficient ceiling or full operator-norm budget. Five runs stop under loss tolerances and three reach evaluation caps. No global optimality is certified. These are four-column common-phase diagnostics on the full register. Both selected fits have basis-state trace errors far above 2^{-16} at the sampled endpoint. The radius is a sampled full centered norm of dimensionless H ; dividing by it gives the same endpoint after that much unit-norm model time, up to centering phase. It is not a certified exact spectrum. Alternatively $B_1 = \sum_k |c_k|$ supplies the exact conservative budget $\|H/B_1\| \leq 1$, with deadline B_1 . No favorable accuracy-latency or physical-resource comparison is promoted from these unsuccessful fits.

The exact frozen Hamiltonians have Gaussian-rational matrix entries. For any selected logical input and its required

output, static dynamics conserve the expectation of H^2 . Trace duality therefore gives

$$D_{\text{tr}}(e^{-itH}|e\rangle, |f\rangle) \geq \frac{|\langle e, H^2 e \rangle - \langle f, H^2 f \rangle|}{2B_1^2} \quad (14)$$

for every deadline. The endpoint-only best fit has a floor strictly exceeding $1/2048$ on input $|10\rangle$. The penalty batch best has a floor strictly exceeding $1/8192$ on

$$|\xi\rangle = (|01\rangle + i|11\rangle)/\sqrt{2}. \quad (15)$$

Both exceed $1/65536$; exact fractions are in the audits. This coherent witness tests the admitted input space rather than only four population transfers. Uniform norm rescaling and a scalar energy shift cannot evade an all-time state exclusion. Each witness concerns only its specified frozen generator; other coefficients, spectral branches, and graphs remain unexcluded.

V. SPECTRAL ALIGNMENT BEYOND MOMENT MATCHING

The allowed one-local candidate $H = X_z$ has full norm one, spectrum $\{-1, 1\}$, and $H^2 = I$. Both compressed odd moments vanish and both even moments equal I_4 . Thus every nonnegative compressed moment matches exactly. It also breaks Q_A, Q_O and activates N_z . Yet it conserves both native occupations, so input 01 remains orthogonal to its required native-zero output for every deadline. Its complete error is one on that input. Activation and even all-order moment matching are insufficient.

For distinct eigenvalues, let Π_λ be the true spectral projectors. Exact admission is equivalent to

$$e^{-it\lambda} \Pi_\lambda E = e^{i\phi} \Pi_\lambda W \quad \text{for every } \lambda. \quad (16)$$

Necessity follows by projection; sufficiency follows by summing the complete projectors. This condition retains orientation inside each eigenspace and one common phase across the four columns.

Define the spectral trace overlap

$$S = \sum_\lambda |\text{Tr}(W^\dagger \Pi_\lambda E)| \leq 4. \quad (17)$$

Proposition 1 (All-time complete-channel spectral witness). *For every deadline, the half-diamond distance between the two isometric admission channels satisfies*

$$\frac{1}{2} \|\mathcal{U}_t - \mathcal{W}\|_\diamond \geq \sqrt{1 - S^2/16}. \quad (18)$$

Proof. Use the maximally entangled logical/reference input of dimension four. Its two output states are pure, with overlap $\text{Tr}(W^\dagger e^{-itH} E)/4$. Expanding into projectors and applying the triangle inequality bounds its modulus by $S/4$. The pure-state trace distance is the displayed radical and lower-bounds half-diamond distance. Global phase does not change it. $\square$

For X_z , $S = 1$, giving the relaxed reference witness $\sqrt{15}/4$, while the basis witness already gives one. Moreover $S = 4$ forces equality in the matrix and scalar Cauchy-Schwarz inequalities: each $\Pi_\lambda W$ must equal $\Pi_\lambda E$ up to a scalar phase shared by all four columns. Those eigenspace phases must still fit one finite deadline. This separates orientation from phase timing; neither is closed by compressed moments alone.

A. Exact enclosure for the frozen fits

Treat the floating eigendecomposition (V, Λ) only as exact binary-rational data. Compute Gaussian-integer Gram and eigen residuals, with rational upper enclosures

$$\delta \geq \|V^\dagger V - I\|_F, \quad \rho \geq \|HV - V\Lambda\|_F. \quad (19)$$

The polar unitary Q satisfies $\|Q - V\| \leq \delta$ for $\delta < 1$. Thus, with $B_1 = \sum |c_k|$,

$$\|H - Q\Lambda Q^\dagger\| \leq \eta := \rho + (B_1 + \|\Lambda\|)\delta. \quad (20)$$

Let γ be the smallest exact binary-rational grid gap. The receipts verify $\gamma > 2\eta$. Eigenvalue perturbation then identifies one true rank-one projector per grid point. Projecting the true eigenvector equation onto other grid eigenspaces bounds the projector displacement by $\eta/(\gamma - \eta)$. Replacing $Q_n Q_n^\dagger$ by $V_n V_n^\dagger$ adds at most 3δ .

Each four-column trace overlap incurs at most four times this projector debit. Summing the 64 terms yields

$$S \leq S_{\text{up}} := \sum_n |\text{Tr}(W^\dagger V_n V_n^\dagger E)|_{\text{up}} + 256 (\eta/(\gamma - \eta) + 3\delta). \quad (21)$$

Every square root is enclosed by exact integer square-root arithmetic on an 80-bit dyadic grid. Both certified $S_{\text{up}} < 4$ and $1 - (S_{\text{up}}/4)^2 > (1/8)^2$. Hence both frozen generators have all-time complete channel error strictly greater than $1/8$, including under uniform norm rescaling. Diagnostic radical values are 0.246049 and 0.191983. These are lower bounds, not exact channel distances. Neither certificate excludes other coefficients or a different spectral branch.

VI. VERIFICATION AND NEXT GATE

This release records 18 exact basis and decomposition checks, two nine-check Gaussian-rational moment audits, a 16-check counterexample, and two 73-check spectral enclosures: 198 exact checks. Numerical evidence comprises eight fits, 392 gradient-column comparisons and two directional comparisons. Independent review and theorem-level novelty comparison remain pending; physical evidence promotion is zero.

The next mathematical gate is spectral-subspace alignment on the same planar graph, followed by phase compatibility at a finite normalized deadline. Moment constraints remain useful prefilters; a soft penalty cannot certify closure. Near-degenerate gaps and long phase-fitting times must be charged rather than treated as free solutions. An exact or interval-certified candidate remains the target. The present finite search is not a proof that the planar graph must change or that a multi-excitation carrier is necessary. Native extraction, static concatenation, irreversible retention, supervising-clock refresh and thermodynamic supply remain separate open obligations.

REFERENCES

- [1] B. W. Mayes, "Moment-constrained symmetry breaking for autonomous joint qubit admission," MQM development note v0.1, Oct. 5, 2026, unpublished. Frozen parent identified in INPUT_ANCHORS.json.
- [2] L. Innocenti, L. Banchi, A. Ferraro, S. Bose, and M. Paternostro, "Supervised learning of time-independent Hamiltonians for gate design," arXiv:1803.07119, 2018, doi: 10.1088/1367-2630/ab8aaf.